RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie
....
MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR
....
COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN HIGHER NATIONAL DIPLOMA EXAM (HND)
....

REPUBLIC OF CAMEROON
Peace-Work-Fatherland
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MINISTRY OF HIGHER EDUCATION
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NATIONAL COMMISSION FOR THE ORGANISATION OF HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM
....

National Exam of Higher National Diploma-New program – 2020 Session

Spécialty/option : NWS-SWE-EDM
Paper : Discrete Mathematics
Duration : 4 hours

Credit : 4

Instructions: Answer all questions. You are authorized to use only non-programmable calculator.

SECTION A: MCQS (20 MARKS)

1) **Suppose that the mean of the sampling distribution for the difference in two sample proportions is 0.** This tells us that:
   A. The two population proportions are both 0.
   B. The two population proportions are equal to each other.
   C. The two sample proportions are both 0.
   D. The two sample proportions are equal to each other.
2) **If A and B are independent events (both with probability greater than 0) then which of the following statements must be true?**
   A. P(A) + P(B) = 1
   B. P(A and B) = P(A)P(B)
   C. P(A and B) = 0
   D. P(A and B) = P(A) + P(B)
3) **A sales person makes "cold calls" trying to sell a product by phone and is successful on each call with probability 1/50. Whether or not he is successful is independent from one call to the next. If he calls 50 people, the number of successful calls is:**
   A. exactly 1, since he called 50 people and the probability of success is 1/50 each time.
   B. at most 1, because once he has been successful he can't be successful again in the 50 calls.
   C. a binomial random variable.
   D. equally likely to be 0, 1 or 2.
4) **The expected value of a random variable is**
   A. always computed as np.
   B. the value that has the highest probability of occurring.
   C. always one of the possible values for the random variable.
   D. the mean value over an infinite number of observations of the variable.

5) **Suppose that a 95% confidence interval for the proportion of men over 60 who have high blood pressure is .30 to .40. Which of the following is the best interpretation of this information?**
   A. 95% of the men in the sample have blood pressure that is between 30% and 40% too high.
   B. 95% of the men in the population have blood pressure that is between 30% and 40% too high.

C. We can be fairly confident (about 95%) that the proportion of men in the sample who have high blood pressure is between .30 and .40.

D. We can be fairly confident (about 95%) that the proportion of men in the population who have high blood pressure is between .30 and .40.

6) **All of the following are ways to show that A and B are mutually exclusive events except one of them. Which one does not show that A and B are mutually exclusive?**

A. $P(A|B) = 0$ B. $P(A \text{ and } B) = P(A)P(B)$

C. $P(A \text{ n } B) = 0$ D. $P(A \text{ or } B) = P(A) + P(B)$

7) **You buy coffee at a kiosk at random times. The following table gives the probability distribution for X = number of customers in line when you show up (not including you):**

K	0	1	2	3	4
P(X = k)	0.15	0.20	0.40	0.20	0.05

The probability that there will be at least two people in line when you show up is

A. 0.25 B. 0.40 C. 0.65 D. 0.75

Use this information to answer problems 8 and 9. x lies in the first quadrant and y lies in the second quadrant. Sin $x = \frac{4}{5}$, and cos $y = -\frac{8}{17}$.

8) **The exact value of cos $(x+y)$ is**

A. $\frac{13}{85}$ B. $-\frac{36}{85}$ C. $\frac{36}{85}$ D. $\frac{84}{85}$

9) **The exact value of sin $2x$ is**

A. $\frac{8}{5}$ B. $\frac{24}{25}$ C. $\frac{9}{25}$ D. $\frac{3}{5}$

10) **Which of the following is *not* a solution to sin $2x$ + cos x = 0 over the interval $[0, 2\pi)$?**

A. $\frac{\pi}{2}$ B. $\frac{5\pi}{6}$ C. $\frac{7\pi}{6}$ D. $\frac{3\pi}{2}$

11) **A class contains 8 boys and 7 girls. The teacher selects 3 of the children at random and without replacement. The probability that the number of boys selected exceeds the number of girls selected is**

A. $\frac{512}{3375}$ B. $\frac{28}{65}$ C. $\frac{8}{15}$ D. $\frac{1856}{3375}$

12) **What are all values of k for which $\int_{-3}^{k} x^2 dx = 0$**

A. −3 B. 3 C. −3 and 3 D. −3, 0, and 3

13) Find the value(s) of $\frac{dy}{dx}$ of $x^2y+y^2=5$ at $y=1$.

A. $-\frac{2}{3}$ only B. $\frac{2}{3}$ only C. $\pm\frac{2}{3}$ D. $\pm\frac{3}{2}$

14) Find the average rate of change of the function f(x) on [0, 2] if $f(x)=2x^2-2$

A. –4 B. 0 C. 1 D. 4

15) The domain of the function $f(x) = \sqrt{1-x}\ \ln x$ is

A. $(0, +\infty)$ B. $(0, 1]$ C. $(-\infty, 1)$ D. $[0, 1]$

16) The exact value of $\lim_{x\to 0}\frac{\sqrt{3+x}-\sqrt{3}}{x}$ is

A. $\sqrt{3}$ B. 0 C. $\frac{1}{2\sqrt{3}}$ D. the limit does not exist

17) Which of the following is an equation of the tangent to the curve $y = 2x\sin x$ at the point $(\frac{\pi}{2}, \pi)$?

A. $y = 2x + 2\pi$ B. $y = 2x$ C. $y = -2x$ D. $y = -2x + 2\pi$

18) The exact value of $\lim_{x\to\infty}\frac{x+2}{9x^2+1}$ is

A. 0 B. $\frac{1}{9}$ C. $\frac{2}{9}$ D. ∞

19) Which of the following is an even function of t.?

A. $3t^2$ B. t^2-4t C. $\sin 2t + 3t$ D. t^3+6

20) "A periodic function" is given by a function which

A. has a period $T = 2\pi$ B. has a period $T = \pi$

C. satisfies $f(t + T) = - f(t)$ D. satisfies $f(t + T) = f(t)$

21) For $y = \sin^2 x + \cos^2 x$, $y' =$

A. $2\sin x - 2\cos x$ B. $2\sin x\cos x$ C. $4\sin x\cos x$ D. 0

22) If $y^3 = x^2$, then $\frac{dy}{dx} =$

A. $\frac{2x}{y^3}$ B. $\frac{2x}{3y^2}$ C. $\frac{x^2}{3y^2}$ D. $\frac{x^2}{3y}$

23) $\operatorname{cosec}(-\frac{4}{3}\pi) =$

A. 2 B. $\frac{\sqrt{3}}{2}$ C. $\frac{2\sqrt{3}}{3}$ D. $-\frac{2\sqrt{3}}{3}$

24) Find a positive value c, for x, that satisfies the conclusion of the Mean Value Theorem for Derivatives for $f(x) = 3x^2 - 5x + 1$ on the interval [2, 5].

A. 1 B. $\frac{7}{2}$ C. 3 D. $\frac{23}{6}$

25) At what value of x does the function $f(x)=\dfrac{(x+1)^2}{x^2-1}$ have a removable discontinuity?

A. 1 B. 3 C. 2 D. -1

26) Suppose g is a function defined on the interval [0, 5] such that $\lim_{x\to 2} g(x)=-1$, the $\lim_{x\to 2}[g(x)]^3$ is

A. 8 B. $\frac{1}{8}$ C. -1 D. $\frac{1}{3}$

27) If $f(a)=f(b)=0$ and *f(x)* is continuous on [a, b], then

A. *f(x)* must be identically zero,
B. *f'(x)* may be different from zero for all *x* on (a, b).
C. There exists at least one number c, $a < c < b$, such that $f'(c) = 0$,
D. *f'(x)* must exist for every *x* on (a, b).

28) A medical treatment has a success rate of 0.8. Two patients will be treated with this

treatment. Assuming the results are independent for the two patients, what is the probability

that neither one of them will be successfully cured?

A. 0.5 B. 0.36 C. 0.2 D. 0.04

29) Suppose that the probability of event A is 0.2 and the probability of event B is 0.4. Also, suppose that the two events are independent. Then P(A|B) is:

A. P(A) = 0.2 B. P(A)/P(B) = 0.2/0.4 = ½

C. P(A) × P(B) = (0.2)(0.4) = 0.08 D. None of the above.

30) A specific range of numbers within which a population mean should lie is

A. the range. B. the confidence coefficient.
C. the confidence interval. D. the confidence level.

31) The sample standard deviation using the data

X: 4 5 2 6 4 3 4 is

A. 10 B. 1.66 C. 1.20 D. 1

32) A sample of 40 cows is drawn to estimate the mean weight of a large herd of cattle. If the standard deviation of the sample is 96 kg, what is the maximum error in a 90% confidence interval estimate?

A. 25 kg B. 158 kg C. 58 kg D. 30 kg

33) Find the Laplace transform of f(t) = 3.

A. $\frac{3}{s}$ B. $\frac{1}{s}$ C. 3 D. 3s

34) Find the $L^{-1}(\frac{1}{(s+2)^4})$.

A) $e^{-2t}\times 3$ B) $e^{-2t}\times\frac{t^3}{3}$ C) $e^{-2t}\times\frac{t^3}{6}$ D) $e^{-2t}\times\frac{t^2}{6}$

35) The coefficient of the term in x^2 in the Maclaurin expansion of $In(1-2x)$ is

A) $\frac{1}{2}$ B) 1 C) -2 D) 2

36) State the interval of convergence the function

$f(x) = x^2 \ln(1-2x)$

A. $I = (1, 2)$ B. $I = [-\frac{1}{2}, \frac{1}{2})$ C. $I = [1, 2]$ D. $I = [-\frac{1}{2}, \frac{1}{2}]$

37) $f(x) = \sqrt{4+x}$ **as a Maclaurin series, State it's radius of convergence.**

A. $R = 2$ B. $R = 0$ C. $R = 3$ D. $R = 1$

38) $\sec^2 x - \tan^2 x$ is identical to which of the following:

A. 1 B. $\sin^2 x - \cos^2 x$. $\cos^2 x - \sin^2 x$ D. $1 - \tan^2 x$

39) What is the magnitude of the vector $\langle 6, 15 \rangle$?

A. $\sqrt{21}$ B. 261 C. 21 D. $\sqrt{261}$

40) What are the first five terms of the sequence defined as

$a(1) = 3$

$a(n+1) = a(n) - 4$, for $n \geq 1$?

A. $-3, -2, -1, 0, 1$ B. $-1, -5, -9, -13, -17$

C. $3, -1, -5, -9, -13$ D. $3, -1, 0, 1, 2$

SECTION B: STRUCTURAL (80 MARKS)

1. Analysis (30 Marks)

1.1. Evaluate $f_x(x, y)$ *and* $f_y(x, y)$ for the following functions.

(i) $f(x, y) = x^3 + 5xy + 2y - 2$

(ii) $f(x, y) = 8 + cos2x - xy^2$. **(2+ 2 marks)**

1.2. Evaluate $\int_0^1 \int_0^2 \int_0^2 x^2 yz \, dz \, dy \, dx$ · **(4 marks)**

1.3. a) Evaluate the Laplace transforms of $3t^2 + \cos 2t$.

b) Solve the following initial value problem by using the Laplace transform

$y' + 2y = 3$ $y(0) = 0$ **(3+ 5 marks)**

1.4. Sketch the triangular wave

$$f(t) = \begin{cases} t, & 0 < t < \pi \\ -t, & -\pi < t < 0 \end{cases}$$

$f(t) = f(t + 2\pi)$, for $-5\pi < t < 5\pi$

Is the function odd or even? Show that the corresponding Fourier series is

$f(t)$ = Erreur !– Erreur ! **(6 marks)**

1.5. a) Solve the differential equation

$$\left(y^2 - xy\right)dx + x^2 dy = 0.$$

b) Find the particular solution of the differential equation $2y''-3y'+2y=0$ which satisfies the boundary condition $y=0$ and $y'=2$ when $x=0$. **(4+4 marks)**

2. Statistics (30 Marks)

2.1 A national examination in Mathematics was taken by 8479 candidates and the result is summarized in the grouped frequency distribution below.

Marks	0-10	11-20	21-30	31-40	41-50	51-60	61-70	71-80	81-90	91-100
No of candidates	46	216	822	1057	1492	1683	1522	1011	522	108

It is decided that 62.2% of the candidates should pass.

a) Calculate the necessary pass mark, regarding the marks as a continuous variable.

b) Draw a cumulative frequency graph and hence estimate the range of marks obtained by the central 80% of candidates.

c) If the pass mark has been fixed for 55%, how many candidates would pass? **(3+ 7+ 2 marks)**

2.2 The probability density function for a continuous random variable X is

$$f(x) = \begin{cases} a + bx^2, & 0 \le x \le 1 \\ 0, & otherwise \end{cases}$$

where a, b are some constants. Find

a) a, b if $E(X) = \dfrac{3}{5}$

b) $Var(X)$. **(4+ 4 marks)**

2.3 The number of customers arriving at a Bank is Poisson distributed with a mean, 4 customers/per minute.

a) Within 2 minutes, what is the probability that there are 3 customers?

b) What is the probability density function for the time between the arrival of the next customer? **(5+5 marks)**

3. Probability (20 Marks)

3.1. a) State Baye's Theorem

b) Suppose a certain disease has an incidence rate of 0.1% (that is, it afflicts 0.1% of the population). A test has been devised to detect this disease. The test does not produce false negatives (that is, anyone who has the disease will test positive for it), but the false positive rate is 5% (that is, about 5% of people who take the test will test positive, even though they do not have the disease). Suppose a randomly selected person takes the test and tests positive. Using Baye's Theorem find the probability that this person actually has the disease? **(2+ 4 marks)**

3.2. In a random sample of HND students 50% indicated they are business majors, 40% engineering majors, and 10% other majors. Of the business majors, 60% were female; whereas, 30% of engineering majors were females. Finally, 20% of the other majors were female. Given that a person is female, what is the probability that she is an engineering major? **(5 marks)**

3.3. Given that $P(A \cup B) = 0.7$ *and* $P(A \cup B') = 0.9$, find P(A). **(4 marks)**

3.4. A bloods test indicates the presence of a particular disease 95% of the time when the disease is actually present. The same test indicates the presence of the disease 0.5% of the time when the disease is not present. One percent of the population actually has the disease. Calculate the probability that the person has the disease given that the test indicates the presence of the disease. **(5 marks)**